No Unbounded Convex W-Hyperbolic Set Has the UAFPP

Automated Math Discovery Smoke Test

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Source and Open Problem

This packet concerns the open problem on page 44 of:

> Laurentiu Leustean, *Proof mining in metric fixed point theory and ergodic theory*, arXiv:0903.1529, Oberwolfach Preprints OWP 2009-05.

The same UAFPP problem is also the final open problem of the original Kohlenbach–Leustean paper on approximate fixed points in product spaces. The source asks whether any unbounded convex subset of a W-hyperbolic space can have the uniform approximate fixed point property for all nonexpansive self-maps.

> $T : C \to C$, $T(z) = y$ for all $z \in C$. Then $d(x, Tx) = d(x, y) > 2D_1 + 1$ which contradicts (69).
>
> We conclude this subsection with an **open problem**:
>
> *Are there unbounded convex subsets C of some W-hyperbolic space which have the UAFPP for all nonexpansive mappings $T : C \to C$?*

Definitions

We use the standard notation from the source. A W-hyperbolic space is a metric space (X, d) equipped with a convexity map

$$W : X \times X \times [0, 1] \to X.$$

The only axiom needed below is the arc-length axiom

$$d\big(W(x, y, \lambda), W(x, y, \mu)\big) = |\lambda - \mu|\, d(x, y) \qquad (\lambda, \mu \in [0, 1]).$$

A subset $C \subseteq X$ is convex if $W(x, y, \lambda) \in C$ for all $x, y \in C$ and $\lambda \in [0, 1]$.

Definition 1 (UAFPP)**.** *Let C be a metric space and let $\mathcal{F}$ be a class of maps $T : C \to C$. We say that C has the uniform approximate fixed point property for $\mathcal{F}$ if for every $\varepsilon > 0$ and every $b > 0$ there is $D > 0$ such that, for every $x \in C$ and every $T \in \mathcal{F}$,*

$$d(x, Tx) \le b \quad \Longrightarrow \quad \exists x^* \in C \ \big(d(x, x^*) \le D \text{ and } d(x^*, Tx^*) < \varepsilon\big).$$

Proof Intuition

The source already observes that a stronger, basepoint-free uniform property would force boundedness by testing constant maps. UAFPP avoids that simple obstruction because it only asks for nearby approximate fixed points when the chosen base point is moved by at most b.

The missing obstruction is still one-dimensional. From a base point a choose a very long geodesic segment inside the unbounded convex set. Collapse every point of C to this segment according to its distance from a, then move one unit farther along the segment, stopping only at the far endpoint. This radial map is nonexpansive because both the distance-to-a function and the arc-length parametrization of the segment are 1-Lipschitz. Near a, it increases the distance from a by exactly one, so the reverse triangle inequality forces displacement at least one throughout any prescribed bounded neighborhood of a.

Main Result

Theorem 1. *Let C be a nonempty unbounded convex subset of a W-hyperbolic space (X, d, W). Then C does not have the UAFPP for the class of all nonexpansive self-mappings $T : C \to C$.*

Consequently, the open problem in arXiv:0903.1529 has a negative answer: there are no unbounded convex subsets of W-hyperbolic spaces with the UAFPP for all nonexpansive self-maps.

Proof. Fix $a \in C$. Suppose, toward a contradiction, that C has the UAFPP for all nonexpansive self-maps. Apply the definition with $\varepsilon = 1$ and $b = 1$, and let $D > 0$ be the corresponding radius.

Since C is unbounded, choose $y \in C$ such that

$$L := d(a, y) > D + 1.$$

For $t \in [0, L]$, define the geodesic segment

$$\gamma(t) := W(a, y, t/L).$$

By convexity of C, $\gamma([0, L]) \subseteq C$, and by the arc-length axiom,

$$d(\gamma(s), \gamma(t)) = |s - t| \qquad (s, t \in [0, L]).$$

Define $h : [0, \infty) \to [0, L]$ by

$$h(r) := \min\{r + 1, L\},$$

and define $T : C \to C$ by

$$Tz := \gamma\big(h(d(a, z))\big).$$

The function h is 1-Lipschitz, and $z \mapsto d(a, z)$ is 1-Lipschitz. Therefore, for all $z, z' \in C$,

$$\begin{aligned} d(Tz, Tz') &= \big|h(d(a, z)) - h(d(a, z'))\big| \\ &\leq |d(a, z) - d(a, z')| \leq d(z, z'). \end{aligned}$$

Thus T is nonexpansive. Moreover,

$$d(a, Ta) = d(\gamma(0), \gamma(1)) = 1,$$

so the UAFPP hypothesis applies at the base point a.

We now show that no point in the D-neighborhood of a is a 1-fixed point of T. Let $z \in C$ satisfy $d(a, z) \leq D$. Since $D + 1 < L$, we have $h(d(a, z)) = d(a, z) + 1$. Hence

$$d(a, Tz) = d(a, z) + 1.$$

By the reverse triangle inequality,

$$d(z, Tz) \geq |d(a, Tz) - d(a, z)| = 1.$$

Thus $d(z, Tz) < 1$ fails for every $z \in C$ with $d(a, z) \leq D$.

This contradicts the UAFPP conclusion for $\varepsilon = 1$, $b = 1$, the point a, and the nonexpansive map T. Therefore no such unbounded convex set C has the UAFPP. □

Verification and Novelty Notes

The proof is purely metric and uses only the arc-length axiom of W-hyperbolic spaces, convexity of C, and the reverse triangle inequality. No computation is used in the proof.

Before promotion, the run indexes were searched for ‘0903.1529‘, proof mining, metric fixed point theory, UAFPP, approximate fixed point property, and related fixed-point keywords. No exact prior packet for this source problem was found. Nearby registry hits concerned other fixed-point or metastability targets.

A bounded web search using the phrases ‘uniform approximate fixed point property‘, ‘unbounded convex subsets‘, ‘W-hyperbolic‘, ‘UAFPP‘, and the exact open-problem wording found the source arXiv survey and the original Kohlenbach–Leustean article, both presenting the problem as open, but did not find a later solution.

Human Review Recommendation

Review as a likely-valid full solution. The main check is whether the arc-length property quoted above is indeed part of the source's definition of W-hyperbolic space; it is the standard axiom used in this literature. Once that is confirmed, the nonexpansiveness and contradiction are immediate.

References

[1] L. Leustean, *Proof mining in metric fixed point theory and ergodic theory*, arXiv:0903.1529, Oberwolfach Preprints OWP 2009-05, 2009.

[2] U. Kohlenbach and L. Leustean, *The approximate fixed point property in product spaces*, Nonlinear Analysis: Theory, Methods & Applications 66 (2007), 806–818.